

Weekly Report for BSc Project

Student RUG ID:	2312553
Student IIT ID:	20220580
Student name	Bopage Rivindu Prabha Akasha
Course	BSc Artificial Intelligence and Data Science
Module	CM4609 - Individual Research Project
Project Topic	Dynamic Narrative Personalization in Games Using AI
Week commencing on dd / mmm / yyyy	23/03/2026
Meeting with the supervisor	Yes Final submission review and logbook preparation discussed.
Description of progress this week	Completed final proofreading and formatting of project report. Prepared logbook for submission. Reviewed all chapters for consistency and ensured references were correctly formatted.
Plan for next week	Prepare for viva.
Date of report	23/03/2026
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Module	CM4609 - Individual Research Project
Project Topic	Dynamic Narrative Personalization in Games Using AI
Week commencing on dd / mmm / yyyy	16/03/2026
Meeting with the supervisor	Yes Feedback on thesis discussion chapter and limitations section. Reviewed evaluation methodology.
Description of progress this week	Wrote discussion chapter covering interpretation of quantitative and qualitative results. Analysed system strengths and limitations including LLM latency and dependency on telemetry quality. Drafted conclusion chapter.
Plan for next week	Final proofreading, formatting, and submission of project report and logbook.
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Module	CM4609 - Individual Research Project
Project Topic	Dynamic Narrative Personalization in Games Using AI
Week commencing on dd / mmm / yyyy	09/03/2026
Meeting with the supervisor	No
Description of progress this week	Completed end-to-end integration testing of full pipeline. Wrote analysis scripts for GEQ Cronbach's alpha computation and NPS-3 scoring. Prepared semi-structured interview guide for qualitative evaluation.
Plan for next week	Write discussion chapter and finalise results section.
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Module	CM4609 - Individual Research Project
Project Topic	Dynamic Narrative Personalization in Games Using AI
Week commencing on dd / mmm / yyyy	02/03/2026
Meeting with the supervisor	No
Description of progress this week	Implemented evaluation instruments including GEQ Flow and Immersion subscales, NPS-3, and miniPXI questionnaires. Conducted initial dry-run test of full system with testbed game. Began writing results chapter.
Plan for next week	Complete integration testing, write analysis scripts, and prepare interview guide.
Date of report	02/03/2026
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Project Topic	Dynamic Narrative Personalization in Games Using AI
Week commencing on dd / mmm / yyyy	23/02/2026
Meeting with the supervisor	Yes Interim demo of the full pipeline. Feedback received on evaluation depth and quantitative analysis approach.
Description of progress this week	Demonstrated end-to-end system: telemetry stream to player model to narrative generation to Unity content injection. Received feedback on strengthening the evaluation design. Began preparing evaluation instruments.
Plan for next week	Implement evaluation questionnaires and begin user study preparation.
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Week commencing on dd / mmm / yyyy	16/02/2026
Meeting with the supervisor	No
Description of progress this week	Completed Unity plugin components: NarrativeManager polling loop, ContentInjector event system, and PluginConfig ScriptableObject. Wrote EditMode unit tests for TelemetryBatch. Added null guard and OnDestroy cleanup to plugin.
Plan for next week	Full system demonstration and supervisor feedback session.
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Week commencing on dd / mmm / yyyy	09/02/2026
Meeting with the supervisor	Yes Feedback on Unity plugin architecture and telemetry schema design. Discussed integration strategy between Unity and FastAPI backend.
Description of progress this week	Implemented PlayerDataLogger in Unity to stream 23 behavioural signals every five seconds. Integrated Unity plugin with FastAPI backend via WebSocket. Performed initial testing on testbed game scenes.
Plan for next week	Complete NarrativeManager and ContentInjector components.
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Meeting with the supervisor	No
Description of progress this week	Implemented Gymnasium MDP environment with Flow-based reward function. Integrated PPO agent using Stable-Baselines3 for narrative action selection across eight action types. Built narrative REST endpoint and completed backend integration.
Plan for next week	Begin Unity plugin implementation.
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Week commencing on dd / mmm / yyyy	26/01/2026
Meeting with the supervisor	No
Description of progress this week	Built RAG retriever using ChromaDB and sentence-transformers (all-MiniLM-L6-v2). Implemented episodic session memory store and dynamic narrative prompt builder. Integrated Ollama LLM client for local inference.
Plan for next week	Implement RL environment and PPO agent for narrative action selection.
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Module	CM4609 - Individual Research Project
Project Topic	Dynamic Narrative Personalization in Games Using AI
Week commencing on dd / mmm / yyyy	19/01/2026
Meeting with the supervisor	Yes Mid-implementation check-in. Discussed Hexad profiling approach and derivation of player types from telemetry without questionnaires.
Description of progress this week	Implemented Hexad profiler producing a six-dimensional continuous player type vector. Completed flow classifier using XGBoost. Tested feature extraction pipeline with sample telemetry data.
Plan for next week	Begin narrative pipeline implementation (RAG, memory, prompt builder).
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Module	CM4609 - Individual Research Project
Project Topic	Dynamic Narrative Personalization in Games Using AI
Week commencing on dd / mmm / yyyy	12/01/2026
Meeting with the supervisor	No
Description of progress this week	Implemented WebSocket endpoint for real-time telemetry ingestion. Built feature extractor mapping 23 behavioural signals to an 11-feature player model vector. Completed SQLite telemetry store with session queries.
Plan for next week	Implement Hexad profiler and flow state classifier.
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Module	CM4609 - Individual Research Project
Project Topic	Dynamic Narrative Personalization in Games Using AI
Week commencing on dd / mmm / yyyy	05/01/2026
Meeting with the supervisor	Yes Post-break planning session. Reviewed implementation roadmap and agreed on module delivery order.
Description of progress this week	Resumed work after Christmas break. Configured development environment (Python 3.12, Unity 6 LTS, Ollama). Scaffolded FastAPI backend with Pydantic schemas for telemetry, player model, and narrative.
Plan for next week	Implement SQLite telemetry store and WebSocket endpoint.
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Module	CM4609 - Individual Research Project
Project Topic	Dynamic Narrative Personalization in Games Using AI
Week commencing on dd / mmm / yyyy	08/12/2025
Meeting with the supervisor	Yes Reviewed system design and methodology chapter drafts. Feedback on MDP reward function design and cold-start heuristic approach.
Description of progress this week	Finalised system architecture documentation. Completed methodology chapter draft including MDP state and action space definitions, reward function design, and cold-start heuristic (120-second rule before PPO activates).
Plan for next week	Christmas break. Plan implementation phase for January.
Date of report	08/12/2025
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Module	CM4609 - Individual Research Project
Project Topic	Dynamic Narrative Personalization in Games Using AI
Week commencing on dd / mmm / yyyy	01/12/2025
Meeting with the supervisor	No
Description of progress this week	Designed Gymnasium MDP for PPO agent: defined eight narrative action types, state space comprising Flow classification and Hexad profile, and reward function based on challenge-skill balance. Wrote testbed game design document (Eryndal 2D top-down RPG).
Plan for next week	Finalise system design and methodology chapters. Prepare for Christmas break.
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Module	CM4609 - Individual Research Project
Project Topic	Dynamic Narrative Personalization in Games Using AI
Week commencing on dd / mmm / yyyy	24/11/2025
Meeting with the supervisor	No
Description of progress this week	Refined five-module architecture (telemetry, player modeling, narrative generation, RL adaptation, Unity plugin). Finalised tech stack selection: FastAPI, Ollama, ChromaDB, Stable-Baselines3, SQLite. Mapped data flow between all modules.
Plan for next week	Design RL environment and testbed game.
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Module	CM4609 - Individual Research Project
Project Topic	Dynamic Narrative Personalization in Games Using AI
Week commencing on dd / mmm / yyyy	17/11/2025
Meeting with the supervisor	Yes Architecture review. Feedback on player modeling approach and evaluation design scope. Agreed on between-subjects study design (N=50).
Description of progress this week	Completed literature review chapter draft. Identified six original contributions of the project. Defined evaluation design using GEQ Flow and Immersion subscales, NPS-3, and miniPXI.
Plan for next week	Begin system architecture design and module planning.
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Week commencing on dd / mmm / yyyy	10/11/2025
Meeting with the supervisor	No
Description of progress this week	Reviewed RL approaches for game adaptation including PPO, DQN, and Actor-Critic methods. Decided on PPO via Stable-Baselines3. Read papers on reward shaping for player experience optimization.
Plan for next week	Finalise literature review and define research gap clearly.
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Week commencing on dd / mmm / yyyy	03/11/2025
Meeting with the supervisor	No
Description of progress this week	Surveyed RAG architectures and episodic memory techniques for LLMs. Reviewed ChromaDB and sentence-transformers documentation. Explored Ollama for local LLM inference to avoid API costs and latency.
Plan for next week	Review RL-for-games literature and select adaptation approach.
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Week commencing on dd / mmm / yyyy	27/10/2025
Meeting with the supervisor	Yes Discussed research questions and project scope. Supervisor confirmed focus on closed-loop personalization covering all five modules.
Description of progress this week	Studied Flow theory (Csikszentmihalyi) and the GEQ measurement instrument. Reviewed Hexad player typology (Marczewski) and Yee player motivations. Drafted initial research questions and project objectives.
Plan for next week	Survey LLM and RAG approaches for narrative generation.
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Week commencing on dd / mmm / yyyy	20/10/2025
Meeting with the supervisor	No
Description of progress this week	Read papers on LLM-based NPC dialogue and procedural narrative generation. Reviewed prior work on player modeling approaches including Bartle types, Yee motivations, and Hexad typology. Began annotated bibliography.
Plan for next week	Study Flow theory and formulate research questions.
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Week commencing on dd / mmm / yyyy	13/10/2025
Meeting with the supervisor	No
Description of progress this week	Conducted systematic literature review on narrative personalization in games. Set up reference manager and began building annotated bibliography covering key papers on AI-driven narrative and player experience.
Plan for next week	Review player modeling and LLM-for-games literature.
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Week commencing on dd / mmm / yyyy	06/10/2025
Meeting with the supervisor	Yes Initial supervision meeting. Confirmed project topic, agreed on objectives and scope, and discussed weekly reporting expectations.
Description of progress this week	Confirmed project topic (AI-driven narrative personalization in games). Set up GitHub repository and project directory structure. Began initial background reading on narrative personalization and player-adaptive systems.
Plan for next week	Conduct systematic literature review.
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Week commencing on dd / mmm / yyyy	29/09/2025
Meeting with the supervisor	No
Description of progress this week	Reviewed project brief and module handbook. Explored potential project topics and shortlisted AI-driven narrative personalization in games as the primary focus. Set up initial GitHub repository.
Plan for next week	First supervision meeting and project topic confirmation.
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Week commencing on dd / mmm / yyyy	22/09/2025
Meeting with the supervisor	Yes Module briefing session. Project requirements, assessment criteria, and logbook submission process explained.
Description of progress this week	Attended CM4609 module briefing. Reviewed assessment criteria and logbook requirements. Began exploring potential project topics in the area of AI and games.
Plan for next week	Narrow down project topic and prepare for first supervision meeting.
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